

Historical data aggregation

Setup

```
packages <- c("readxl", "ggplot2", "ggpubr", "reshape2")
invisible(lapply(packages, library, character.only = TRUE))

find_project_root <- function(start = getwd()) {
  path <- normalizePath(start, mustWork = FALSE)
  for (i in seq_len(8)) {
    if (dir.exists(file.path(path, "Code")) && dir.exists(file.path(path, "Data"))) {
      return(path)
    }
    parent <- dirname(path)
    if (identical(parent, path)) break
    path <- parent
  }
  stop("Set LC_DLNM_BASE_DIR to the project root.")
}

base_dir <- Sys.getenv("LC_DLNM_BASE_DIR")
if (!nzchar(base_dir)) base_dir <- find_project_root()

combined_dir <- file.path(base_dir, "Data/Combined_data")
daily_dir <- file.path(base_dir, "Data/UTCI_data/Daily_data")
processing_dir <- file.path(base_dir, "Results/Processed_data")
legacy_output_dir <- file.path(base_dir, "Results/Processed_data/legacy")
data_visualization_figure_dir <- file.path(base_dir, "Figures/Data_visualization")

dir.create(processing_dir, recursive = TRUE, showWarnings = FALSE)
dir.create(legacy_output_dir, recursive = TRUE, showWarnings = FALSE)
dir.create(data_visualization_figure_dir, recursive = TRUE, showWarnings = FALSE)

write_legacy_output <- TRUE

age_names <- c("Y20_64", "Y65_74", "Y75_84", "Y_GE85")
age_group_labels <- c("Y20_64" = "20-64", "Y65_74" = "65-74",
  "Y75_84" = "75-84", "Y_GE85" = "85+")

city_specs <- data.frame(
  display_name = c("Athens", "Lisbon", "Rome"),
  source_col = c("Attiki", "Lisbon", "Roma"),
  legacy_name = c("Attiki", "Lisbon", "Roma"),
  death_col = c(3, 4, 6),
  exposure_col = c(9, 10, 12),
  loess_span = c(0.05, 0.03, 0.03),
  stringsAsFactors = FALSE
)

line_colors <- c("Y20_64" = "#9ecae1", "Y65_74" = "#6baed6",
```

```

      "Y75_84" = "#2171b5", "Y_GE85" = "#084594")

date_limits <- as.Date(c("2015-01-01", "2020-01-01"))

```

Load Combined Data

```

combined <- list(
  Y20_64 = read_xlsx(file.path(combined_dir, "Y20_64_combined.xlsx")),
  Y65_74 = read_xlsx(file.path(combined_dir, "Y65_74_combined.xlsx")),
  Y75_84 = read_xlsx(file.path(combined_dir, "Y75_84_combined.xlsx")),
  Y_GE85 = read_xlsx(file.path(combined_dir, "Y_GE85_combined.xlsx"))
)

```

Mortality Rate And Wave Lists

```

build_rate_matrix <- function(spec) {
  out <- cbind(
    combined$Y20_64[[spec$death_col]] / combined$Y20_64[[spec$exposure_col]],
    combined$Y65_74[[spec$death_col]] / combined$Y65_74[[spec$exposure_col]],
    combined$Y75_84[[spec$death_col]] / combined$Y75_84[[spec$exposure_col]],
    combined$Y_GE85[[spec$death_col]] / combined$Y_GE85[[spec$exposure_col]]
  )
  colnames(out) <- age_names
  out * 52
}

build_wave_data <- function(spec, use_display_names = TRUE) {
  prefix <- if (use_display_names) spec$display_name else spec$legacy_name
  out <- data.frame(
    hot_wave = combined$Y20_64[[paste0(spec$source_col, "_hot_wave3")]],
    cold_wave = combined$Y20_64[[paste0(spec$source_col, "_cold_wave3")]]
  )
  colnames(out) <- c(paste0(prefix, "_hot_wave3"),
    paste0(prefix, "_cold_wave3"))
  out
}

dat_list <- setNames(lapply(seq_len(nrow(city_specs)), function(i) {
  build_rate_matrix(city_specs[i, ])
}), city_specs$display_name)

wave_list <- setNames(lapply(seq_len(nrow(city_specs)), function(i) {
  build_wave_data(city_specs[i, ], use_display_names = TRUE)
}), city_specs$display_name)

legacy_dat_list <- setNames(dat_list, city_specs$legacy_name)
legacy_wave_list <- setNames(lapply(seq_len(nrow(city_specs)), function(i) {
  build_wave_data(city_specs[i, ], use_display_names = FALSE)
}), city_specs$legacy_name)

```

Save Aggregated Data

```

save(
  dat_list, wave_list, legacy_dat_list, legacy_wave_list,

```

```

    file = file.path(processing_dir, "Historical_data.RData")
  )

  if (write_legacy_output) {
    legacy_env <- new.env(parent = emptyenv())
    legacy_env$dat_list <- legacy_dat_list
    legacy_env$wave_list <- legacy_wave_list
    save(list = c("dat_list", "wave_list"),
         file = file.path(legacy_output_dir, "Historical_data.RData"),
         envir = legacy_env)
  }

```

Historical Mortality Rate Plot

```

make_mortality_long <- function(rate_matrix) {
  plot_data <- as.data.frame(rate_matrix)
  plot_data$Time <- seq(date_limits[1], by = "week", length.out = nrow(plot_data))
  reshape2::melt(plot_data, id.vars = "Time",
                 variable.name = "AgeGroup", value.name = "value")
}

mortality_long <- lapply(dat_list, make_mortality_long)
y_min <- min(unlist(lapply(mortality_long, `[[`, "value")), na.rm = TRUE)
y_max <- max(unlist(lapply(mortality_long, `[[`, "value")), na.rm = TRUE)
year_lines <- seq(as.Date("2016-01-01"), as.Date("2019-01-01"), by = "year")

plot_mortality <- function(data, region) {
  ggplot(data, aes(x = Time)) +
    geom_vline(xintercept = year_lines,
              linetype = "dashed",
              color = "gray50",
              linewidth = 0.3) +
    geom_line(aes(y = value, color = AgeGroup), linewidth = 1.2, alpha = 0.8) +
    geom_smooth(aes(y = value, group = AgeGroup),
               method = "loess",
               span = city_specs$loess_span[match(region, city_specs$display_name)],
               linewidth = 0.5,
               se = TRUE,
               color = "red") +
    labs(title = region, x = "Year", y = "Mortality Rate", color = "Age Group") +
    scale_color_manual(name = "Age Group", values = line_colors, labels = age_group_labels) +
    scale_x_date(
      breaks = seq(date_limits[1], date_limits[2], by = "1 year"),
      date_labels = "%y", limits = date_limits, expand = c(0, 0.01)
    ) +
    scale_y_log10(limits = c(y_min, y_max)) +
    theme_minimal() +
    theme(
      legend.position = "none",
      panel.border = element_rect(color = "black", fill = NA, linewidth = 1),
      panel.grid = element_blank(),
      plot.title = element_text(hjust = 0.5),
      text = element_text(size = 12)
    )
}

```

```

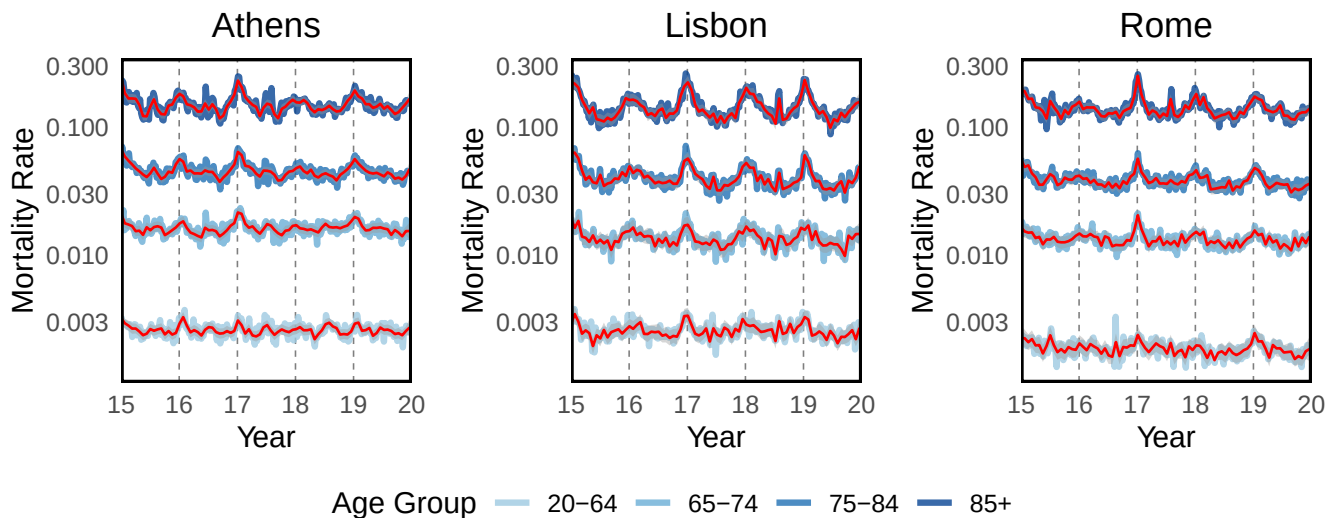
mortality_plots <- lapply(names(mortality_long), function(region) {
  plot_mortality(mortality_long[[region]], region) +
  theme(plot.margin = margin(2, 5, 2, 2, "mm"))
})

Mortality_plot <- ggarrange(
  plotlist = mortality_plots,
  ncol = 3,
  common.legend = TRUE,
  legend = "bottom"
)

ggsave(
  file.path(data_visualization_figure_dir, "Historical_combined.pdf"),
  Mortality_plot, width = 7.5, height = 3, units = "in", dpi = 300
)

print(Mortality_plot)

```



Historical UTCI Plot

```

UTCI_mean <- read_xlsx(file.path(daily_dir, "UTCI_daily_mean.xlsx"))
UTCI_max <- read_xlsx(file.path(daily_dir, "UTCI_daily_max.xlsx"))
UTCI_min <- read_xlsx(file.path(daily_dir, "UTCI_daily_min.xlsx"))

build_utci_data <- function(spec) {
  data.frame(
    Date = as.Date(UTCI_mean$Date),
    Mean = UTCI_mean[[spec$source_col]],
    Min = UTCI_min[[spec$source_col]],
    Max = UTCI_max[[spec$source_col]]
  )
}

utci_list <- setNames(lapply(seq_len(nrow(city_specs)), function(i) {
  build_utci_data(city_specs[i, ])
}), city_specs$display_name)

```

```

plot_utci <- function(data, region) {
  ggplot(data, aes(x = Date)) +
    geom_line(aes(y = Mean, color = "Mean"), linewidth = 0.5) +
    geom_line(aes(y = Min, color = "Minimum"), linewidth = 0.5) +
    geom_line(aes(y = Max, color = "Maximum"), linewidth = 0.5) +
    labs(title = region, x = "Year", y = "UTCI", color = "UTCI") +
    scale_color_manual(
      values = c("Minimum" = "blue", "Mean" = "black", "Maximum" = "red"),
      breaks = c("Minimum", "Mean", "Maximum")
    ) +
    scale_y_continuous(
      breaks = c(-40, -20, 0, 20, 40, 60),
      limits = c(-40, 60), expand = expansion(mult = 0, add = 0)
    ) +
    scale_x_date(
      limits = date_limits, date_breaks = "1 year",
      date_labels = "%Y", expand = c(0, 0.02)
    ) +
    theme_minimal(base_size = 14) +
    theme(
      panel.border = element_rect(color = "black", fill = NA, linewidth = 1),
      panel.grid = element_blank(),
      plot.title = element_text(hjust = 0.5, face = "bold", size = 16),
      legend.position = "bottom",
      plot.margin = margin(t = 1, r = 20, b = 1, l = 5, unit = "pt")
    )
}

UTCI_plot <- ggarrange(
  plotlist = lapply(names(utci_list), function(region) {
    plot_utci(utci_list[[region]], region)
  }),
  ncol = 1,
  nrow = 3,
  common.legend = TRUE,
  legend = "bottom"
)

ggsave(
  file.path(data_visualization_figure_dir, "Historical_UTCI.pdf"),
  UTCI_plot, width = 9, height = 6, units = "in", dpi = 300
)

print(UTCI_plot)

```

